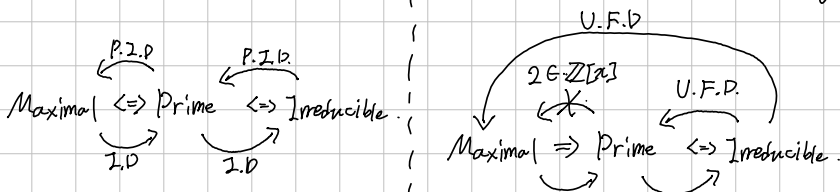
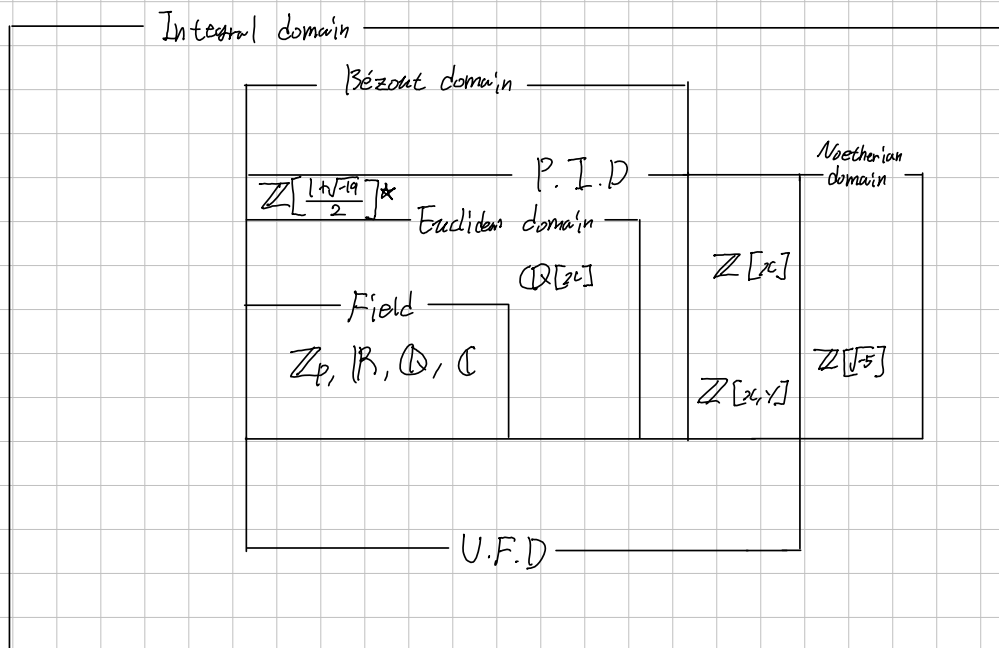


Field $\Rightarrow$ Euclidean Domain $\Rightarrow$ Principal Ideal Domain $\Rightarrow$ Unique Factorization Domain $\subset$ Integral Domain.



P.I.D. $\Leftarrow$ Noether $\Leftrightarrow$ Every ideal are finitely generated $\Leftrightarrow \text{A.C.C.I} \Rightarrow \text{A.C.C.P} \Rightarrow \exists$ factorization $\Leftarrow$ U.F.D.

$\Leftarrow$ Bézout $\Leftrightarrow$ Finitely generated ideal is principal $\Rightarrow$ irreducible = prime $\Rightarrow$! factorization $\Leftarrow$ U.F.D.



Basic languages for generated ideal.

Def. Let R be a Commutative Ring, and $a, b \in R$ with $b \neq 0$.

1. a is called a multiple of b if: there exists $x \in R$ s.t. $a = bx$. Simply $b|a$.
2. A non-zero $d \in R$ is called a Greatest Common divisor of a and b if;

$$d|a \text{ and } d|b \text{ (or, } a, b \in (d)), \quad d'|a \text{ and } d'|b \Rightarrow d'|d. \\ (a, b \in (d') \Rightarrow (d) \subseteq (d'))$$

Note: $b|a \Leftrightarrow a \in (b) \Leftrightarrow (a) \subseteq (b)$.

Directly, $(a, b) = (d)$ implies d is the g.c.d of a and b ,
but Conversely, gcd maybe does not generate the ideal; in $\mathbb{Z}[x]$,

$$(1) \neq (2, x) \text{ but } 1 \text{ is gcd of } 2 \text{ and } x$$

Prop. Let R be a Commutative Ring, and $a, b, c \in R$ be given. Then,

- 1). $(a, b) = (a) + (b)$
- 2). $(c) \cdot [(a) + (b)] = (c) \cdot (a) + (c) \cdot (b) = (ca) + (cb)$.
- 3).

